2.6.6 Dealing with Duplicates

So far, we also assumed that our heap doesn’t hold duplicates: we will have to tackle further challenges if this assumption doesn’t hold true; in particular we will need to figure out the order we use in case there are duplicates. To show why this is important, we will abandon our example for an easier configuration, illustrated in figure 2.11: for the sake of space we will only show priorities in nodes

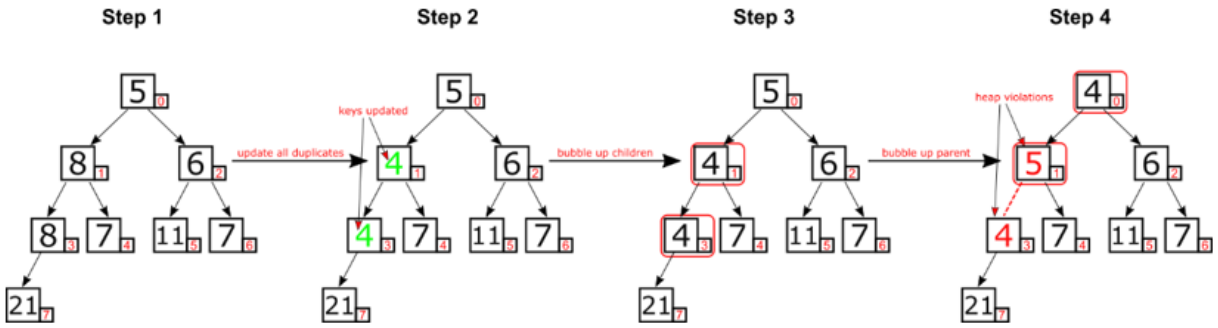


Figure 2.11 Updating an element in a binary min-heap with duplicates.
(Step 1) The initial min-heap. Be aware that, in this example, smaller numbers are considered to have higher priority, because we are using a min-heap.
(Step2) Changing all instances of 8 to 4.
(Step 3) Bubbling up the deepest node updated: it will stop immediately as its parent has the same priority (it just updated together with current node). The nodes involved in the call to `bubbleUp` are highlighted with a red outline.
(Step 4) Bubbling up the other node updated: this time some swaps are actually performed as the new priority is higher than its parent's. The node that was